

Dataset Details

The dataset used in this project was created from industrial workplace safety videos collected from publicly available sources and custom industrial safety footage. The selected videos contain scenes representing workers, personal protective equipment (PPE), machinery, forklifts, hazard zones, welding operations, conveyors, and other workplace safety-related activities.

Since no publicly available dataset matched the specific requirements of industrial worker safety monitoring and PPE compliance detection, a custom object detection dataset was developed for this project.

The videos were processed using OpenCV to extract individual frames. Approximately 683 frames were generated and manually annotated using LabelMe. The annotations were then converted into YOLO object detection format for training the custom YOLO26 model.

Each frame was labeled with bounding boxes and polygons to identify workplace safety-related objects and hazards. The annotated images were divided into training, validation, and testing datasets for model training, evaluation, and performance analysis.

Dataset Information

- Source: Custom Industrial Safety Videos and Publicly Available Industrial Footage
- Dataset Type: Custom Industrial Safety Object Detection Dataset
- Number of Annotated Images: 683
- Annotation Tool: LabelMe
- Annotation Format: JSON (LabelMe)
- Converted Format: YOLO TXT Format
- Data Split: Train, Validation, and Test Sets
- Image Format: JPG
- Label Format: TXT (YOLO Format)

Object Classes

The dataset contains the following safety-related classes:

1. Conveyor
2. Forklift
3. Hazard Zone
4. Helmet
5. Machine Hazard

6. Mask
7. Overhead Hazard
8. Safety Vest
9. Shield
10. Welding Hazard
11. Worker

Dataset Preparation Steps

1. Collected industrial workplace safety videos.
2. Extracted video frames using OpenCV.
3. Generated approximately 683 frames from the collected videos.
4. Manually annotated the extracted frames using LabelMe.
5. Created annotations for workers, PPE equipment, machinery, and hazard-related objects.
6. Converted LabelMe JSON annotations into YOLO TXT format.
7. Organized the dataset into training, validation, and testing folders.
8. Created a data.yaml configuration file for YOLO training.
9. Trained a custom YOLO26 object detection model using the prepared dataset.
10. Evaluated the trained model using validation and testing datasets.

Dataset

https://drive.google.com/drive/folders/1YtR0smOZn0ZOUL93wXvoA37R058eLSVg?usp=s_haring

Extracted Frames

https://drive.google.com/drive/folders/1Q1JxelZbRMLalr1XAp_nP0wTP0sazSXo?usp=drive_link